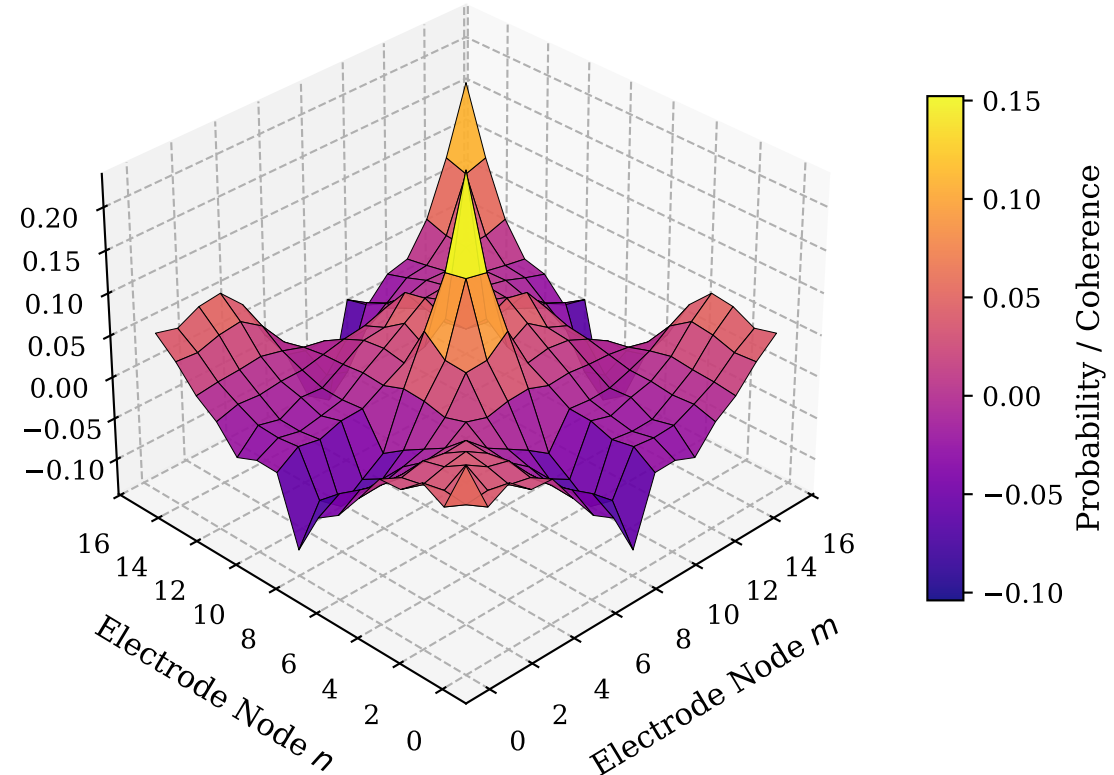
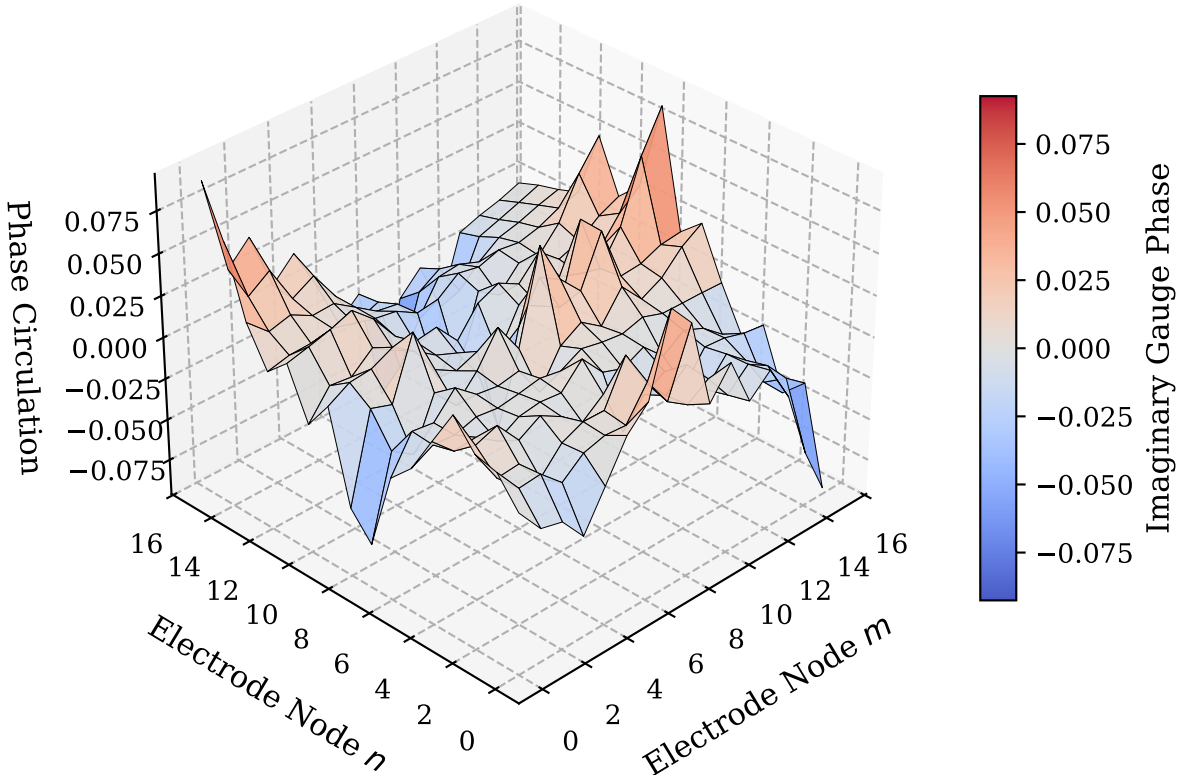
(a) Proposed HQGN: $\mathrm{Re}(\rho(\tau))$ on $\mathcal{H}_{16}$



(b) Proposed HQGN: $\mathrm{Im}(\rho(\tau))$ Chiral Phase Circulation